Vision and Scope Document

for

Fill My Fridge App

Version 1.1 final

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Revision History

Name	Date	Reason For Changes	Version
All members	9/16/24	Chapter 1 - Business Requirements, sections 1.1 - 1.5 Chapter 2 - Scope and Limitations, just section 2.1 - Major Features	1.0 draft
All members	9/24/24	Chapter 1 - Business Requirements, sections 1.6 - 1.7 Chapter 2 - Scope and Limitations - sections 2.2 - 2.4 Chapter 3 - Business Context - sections 3.1 - 3.2	1.0 approved
David Litton	10/07/24	• updating business risks (removed all but one and added 2 more) • removed AS-6 as it we belong in a future section • redefined AS-7 • altered LI 10-12 to EX-1-3 • removed 2 overly specific stake holders	1.1 approved
Ana Knutson	12/12/2024	Removed last section that was unused	1.1 final

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1. Business Requirements

1.1. Background

There are a multitude of cooking assistant apps available today, but there are not any that can do it all. Specifically, recipe collecting with automatic nutrition tracking, meal planning that combines with a smart shopping list which is capable of tracking how much one has for ingredients and their respective expiration dates, does price comparisons between one's top grocery stores, and sets the list up to optimize the route through the selected store, as well as a good cooking mode to show the user the steps they need to take and manage any timers defined in the recipe all in such a way that is easily accessible for the average consumer. Many of these apps have different focuses whether it be solely on nutrition and recipes or grocery lists, while we want to take these key aspects and combine them into one. Below you will find a list of the above apps with specific descriptions about their features and shortcomings we hope to take use to fill the market gap:

- My Fitness Pal is one of the oldest and most well-known apps in this space. It has some of the best nutrition tracking matched to nutrition goals outlined by the user and will recommend recipes shared by its users. However, the app locks off most of these nutrition features behind a premium subscription and only lets free users track calories. Macronutrient and micronutrient tracking as well as meal planning are subscription only. It does not have any features relating to grocery shopping as far as any research was able to find.

- Samsung Food is the new version of one of the arguably best recipe mobile apps, Whisk. It lets the user add or find recipes shared by other users, calculates the nutrition for them, uses them for user-led meal planning, and can use them for creating a shopping list. It also has an ingredient tracker, but from a bit of testing, it does not seem to track the ingredients in any meaningful way. There are no quantities associated with the tracker, nor does it interact with the meal plan to create a grocery list excluding things the user already has. It primarily seems to function as a best-by date tracker and a way to find the ingredient in the kitchen later using a few basic and user defined in-kitchen locations. Samsung Food does a decent job of keeping the essential features free, with the exception nutrition goals. The other premium features are AI recipe personalization, premium meal plans, and an ad-free experience.

- Paprika is an older, established recipe app with a UI to match. It has features for the user to add or find recipes, create a grocery list and a pantry list, a meal planner, and even a "menu" to combine multiple recipes into a meal so one can combine mains and sides together. It doesn't seem to do any nutrition tracking, the grocery list and pantry don't seem to interact in a meaningful way, and the grocery list has to be semi-manually created by adding per recipe instead of generating from the meal plan. The grocery list does do automatic sorting for common aisle groupings for a "store mode". The recipe itself has a non-obvious way to manually add timers, but it is again user-

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led, not automatically from the recipe itself. There is a one time purchase available to remove ads, add a sync feature, and store unlimited recipes.

- Umami seems to be an app designed to replace Paprika considering there is an import feature designed around Paprika. It has all the features of Paprika with the additions of automatic nutrition calculation for recipes, a shareable meal plan and grocery list, and a pseudo-automatic timer system where the user can tap on times in the recipe to start a timer for that step. The number detection is slightly off when given a range, but it is nicer than having to manually input a timer while the input prompt is blocking sight of the time information. It does not seem to track nutrition in a meaningful way, just telling the user the nutrition info of individual recipes. The app itself is also not free. It does have one of the cheaper subscriptions seen so far at just $0.99/month or less if done in bulk, but worth noting there are no free features.

- Cookpad is primarily a recipe app with a higher focus on social features. There are community challenges that can be joined that will show posts where users take pictures of the food after making it. The social features seem to be behind its subscription and the free version of the app is just a basic recipe keeper. There does not seem to be any nutrition info, nor grocery list creation. The challenges appear to be the only meal planning system in place.

- Recipe Keeper has feature parity with Paprika and in addition has a few nice features for sharing, namely the ability to create a recipe book and export it to a PDF to send to another person. The shopping list can also be converted to plaintext easily to send to people who do not use the app. It does limit the number of recipes to 20 until the user makes a one-time purchase to unlock unlimited recipe storage.

- Cronometer is a more health-focused recipe and meal planning app with an emphasis on setting nutrient goals. Much of this is paywalled behind a subscription, but it does let free users set Macronutrient targets and will show how well the user's intended meal plan hits those targets. It does not have any grocery list related features found in testing.

- Pepper is just a recipe keeping app with a high focus on smart imports of recipes from around the web and shared by its users. It has a visually good UI. It is missing anything related to meal planning, nutrition info, or grocery shopping.

- Mealime is another recipe keeper, plus meal planner, plus grocery list maker. It does take a step not seen in any previous recipe app of letting the user add cookware to the recipes so they know what they will need ahead of time. It is very limiting for the free tier, not even allowing user created recipes nor showing any nutrition info. There are also premium only recipes and cooking classes available to subscribers.

- SuperCook is much more focused on the ingredient tracking than any other app tested. The tracker itself is not better than any of the other apps, and in some ways more feature limited, but what it does do is give the user recipes based on the

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ingredients available. It does not seem to have user created recipes without a subscription, but it will browse around the internet for recipes that are strictly limited to the ingredients available. It is more of an index that a keeper.

- Yummly is a recipe keeper, meal planner, grocery list making app sponsored by Whirlpool and is integrated with their smart thermometers. Much of the app is paywalled behind a subscription, including meal planning and "premium" recipes. It features guides on food related topics such as "How to Meal Plan" and "An Introduction to Meal Prepping"

- Tasty is just a recipe keeper and grocery list maker with the added feature of ordering the grocery list through Walmart. It is completely free, and the only ads are for things to add to the grocery list, with most of the monetization likely coming from Walmart. There is also a social feature in the form of a "Community" tab showing blog posts surrounding recipes. There does not appear to be a way for a user to add their own recipes to the app.

- SideChef is another Walmart sponsored recipe app more focused on user defined recipes. It directly maps ingredients to items available for purchase at Walmart. It also has an import system for using OCR (Optical Character Recognition) on images of a physical recipe book. The user can use the app to track ingredients, but testing was inconclusive on if it integrated with the grocery list based on the need to place an order to see if items were automatically transferred to the user's pantry. It features an automatically generated meal-plan based on user preferences. It also appears to be completely free and does not have ads either.

- Grocery - Smart Shopping List is an app by Cooklist, Inc. that offers a wide range of features including price comparison, recipe dictionary and creation, grocery list creation, meal/diet planning, automatic ordering and much more. Maybe their biggest attraction the meal planning being locked behind a pay wall. A feature costing $49.99 annually or $5.99 monthly. Another annoyance our research has discovered is that there is a limited number of ways to register for this app. Either my E-mail the traditional way or by linking your Facebook account. Lastly their UI is cluttered with Icons for dishes, food items, and UI actions making for a distracting and cluttered interface.

1.2. Business Opportunity

The current cooking apps available today are either feature incomplete or locked behind expensive paywalls. By letting the user manage recipes and nutrition goals, create a meal plan based on the recipes and ingredients they are tracking, and use an app created smart grocery list all for free keeping the data locally to the device and only using a subscription system to allow the user to sync across devices and participate in social features such as publishing recipes or sharing meal plans and grocery lists, Fill My Fridge will offer the best cooking app experience available on the market without needing to lock the average user out of essential features. Future

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monetization goals would focus on high quality advertising targeted to a large, receptive userbase by creating sponsored recipes from food and grocery brands with an opt-out feature available to subscribers.

BREQ-1: Simplify the setup process to allow users faster access to our app. Many of the researched app have a complicated set up processes including requiring lots of user information. For example, the app "Cooklist: Pantry & Cooking App" only allows users to choose between manual account creation or by linking a Facebook account. Allowing faster account creation methods or removing the necessity of an account will allow users to more easily access our app and not be deterred by obtrusive account creation.

BREQ-2: Allowing users to document personal recipes as well as upload recipes from common recipe websites will give users the ability to collect a vast library of recipes. We did not observe any other apps give users the option to upload recipes from websites, and this would be a differentiating feature of our app compared to competitors.

BREQ-3: Allow users to plan their meals, keeping a record of their future meals in an intuitive calendar. Hand in hand with grocery shopping is the need to know what it is you plan to make. Having the ability to prerecord what meals you plan to cook makes the shopping experience easier and potentially more cost effective for the user. Clearly linking what they need to get to what food they plan to make. Making the calendar easy to add one recipe across multiple days in a week and scale the number of servings you plan to make on a given day. Keeping track of a user's meals in this manner will prove to be another feature to capitalize on which is not vastly found in the market today.

BREQ-4: Generating grocery lists for the user will lessen the burden on users to memorize everything they need. Creating and organizing lists based on the user's meal plan and referencing what the user already has in their pantry will decrease food waste and over buying at the grocery store. Additionally this will allow for quicker less planned more precise grocery store visits when you can be sure everything is on your list.

BREQ-5: Including nutrition tracking provides users with additional information about the foods they are buying. Giving those using this app an easy way to improve their diets and possibly grocery store habits. The ability to see these statistics will encourage our users to eat better and take stress away from them needing to reading ingredient labels or researching foods online. This will help expand our reach to the users that are already more health focused or are looking to better their eating habits.

BREQ-6: Many grocery stores have different compelling reasons to shop there, some have cheaper produce others cheaper bulk goods. Having the app check nearby grocery stores' prices to find the best deals for the user will increase the appeal of our app. Having an app find the optimal store for each grocery trip will lead to decreased user spending along with

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greater cost awareness for our users if they can see the price of their groceries prior to checkout.

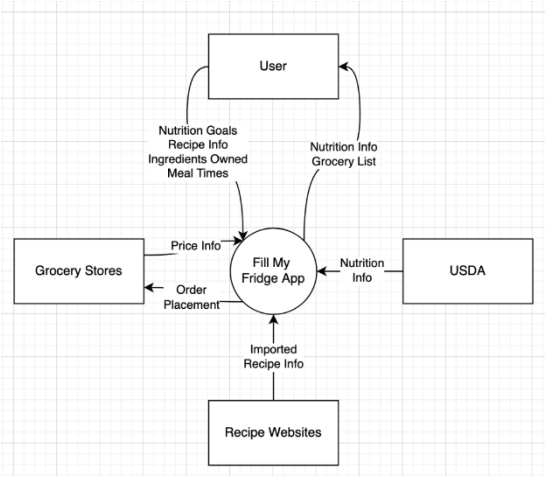


Figure 1: Fill My Fridge App Work Context Diagram

1.3. Business Objectives

BO-1-1: Decrease the average number of clicks to fully access the app compared to competing app upon first release.

BO-1-2: Increase ways to create accounts compared to average number provided by competitors by 20% in the 12 months after initial release.

BO-2-1: In the first version, allow a recipe website to have its recipes be uploaded to our app and move us a step above other applications.

BO-2-2: Include the ability to input personal recipes in the first version to compete with the similar features in other apps.

BO-3-1: Allowing the ability to add one recipe to one or multiple days or meals with limited complexity within 3 months of the first release.

BO-3-2: Show users an overview of their month's meal planning to give users more flexibility than other options upon first release.

BO-4-1: Access store layouts or aisle numbers for items to decrease user time spent in grocery stores by at least 10% within 6 months of the first release.

BO-5-1: Have a full library of food statistics that represents nutrition facts of foundational foods in first release to meet some of the basics of our competitors.

BO-5-2: Include library of branded foods for nutrition facts by the second release to compete with apps such as MyFitnessPal.

BO-6-1: Develop contracts and partnerships with various stores and allow them to advertise products users frequently purchase within 12 months after initial release.

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1.4. Success Metrics

SM-1-1: Comparing average number of click to access full app based on an average of 4.3 clicks from researching competing apps listing in background above, our average must be in the lower 50%, less than 4.3 clicks.

SM-1-2: When polling users for what methods of sign up they expect 75% of user expected account creation methods should be implemented.

SM-2-1: Recipes from AllRecipes recipe website can be uploaded to the app.

SM-2-2: Users can use a basic recipe format to document their personal/outside recipes then access them later.

SM-3-1: Polled users report a 75% satisfactory rating when planning meals using the meal planner.

SM-3-2: When a user views their meal calendar, they are able to see one month worth of planned meals.

SM-4-1: API calls to a database with aisle numbers for major grocery chains comprised of The Kroger Co., Walmart Inc., and Costco Wholesale Corporation.

SM-5-1: API calls to the USDA foundational food database are retrieve correct information 90% of the time.

SM-5-2: API calls to the USDA branded food database are operational and retrieve correct information 80% of the time.

SM-5-3: Visual representation of nutrition facts for foundational foods are represented in the app using a visual diagram.

SM-6-1: Have a partnership with at least one major grocery store chain, such as Walmart Inc., to implement their business and products into our app.

1.5. Vision Statement

For the health conscious home cook, Fill My Fridge offers a better way to set nutrition goals, make meal plans to support those goals, and make sure they know the cheapest price near them to shop at. Unlike the current state of cooking assistant apps, Fill My Fridge uses a local first philosophy so users get the maximum effect of the app for free. For power users who want to create and share recipes, manage a household's meal plan tracking every individual's nutrition goals, or automatically sync data across devices, a subscription service will be available.

1.6. Business Risks

RI-1: Difficulty scraping recipe websites and end up having to scrap this feature. *(Probability = 0.6; Impact = 8)*

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Mitigation Action*: May need to scrap that idea, because if we can only get one website then it could lead to scope creep in trying to get every recipe website implemented. Or, be very specific in only allowing AllRecipes and make it clear to users so they don't have issues.*

RI-2: Amazon Web Service for cloud service for premium members cost too much to attract users. *(Probability = 0.2; Impact = 3)*

Mitigation Action*: Compare Microsoft Azure, Google Cloud Platform (GCP), or other cloud serverless service to find optimal pricing. If this still leads to incompatible pricing see if reverting to an inhouse monolith design would be more cost effective.*

RI-3: Apple decides to make inhouse variant or FMF using exclusive developer tools they have access to corner market on IOS. *(Probability = 0.1; Impact = 10)*

Mitigation Action*: Investigate third party sideloading on IOS. Focus features to be non-Apple ecosystem dependent to separate ourselves from comparison.*

1.7. Business Assumptions and Dependencies

AS-1: People would prefer our app over competitors and make the switch because of the features we have chosen; basically, assuming our app has value

AS-2: Uploading recipes from websites is a feature that people would find interesting and care about

AS-3: There will be dedicated power users who find value in a subscription service

AS-4: Users would know basic cooking terms and don't need cooking tutorials

AS-5: Users have a phone to access the app

AS-6: Users phone will have at least 1 gigabyte of free storage space to handle local storage needs

AS-7: Users will have access to the internet

DE-1: USDA API for foundational foods

DE-2: USDA API for branded foods

DE-3: AllRecipes and other recipe websites aren't too complex to try and scrape

DE-4: Major grocery store chains – need to be able to access their prices and items they have, aisle numbers

DE-5: App Store – can always take our app down so we need them to approve of our app

DE-6: Server – a server to be able to handle when we have subscription users that can share their lists and recipes with others

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2. Scope and Limitations

2.1. Major Features

FE-1: Nutrition tracking, check nutrition values and see personal info stats.

FE-2: Recipe collecting, search, import and create your own recipes.

FE-3: Meal planning, generate meal automatically by nutrition goals or manually where you can save and cancel plans.

FE-4: Pantry tracking, add and track food to your pantry. Add food my scanning barcodes or manually entering food. Track food with expiration dates and nutritional values.

FE-5: Grocery list, automatically create lists from meal plans. Sort lists with custom orders, by type or by templates the user creates.

FE-6: Price comparison, find stores then add them to the comparison.

FE-7: Use easy timers and unit conversions in cooking mode.

FE-8: Store mode shows a simple checklist to check off items you find and automatically adds the items to your pantry.

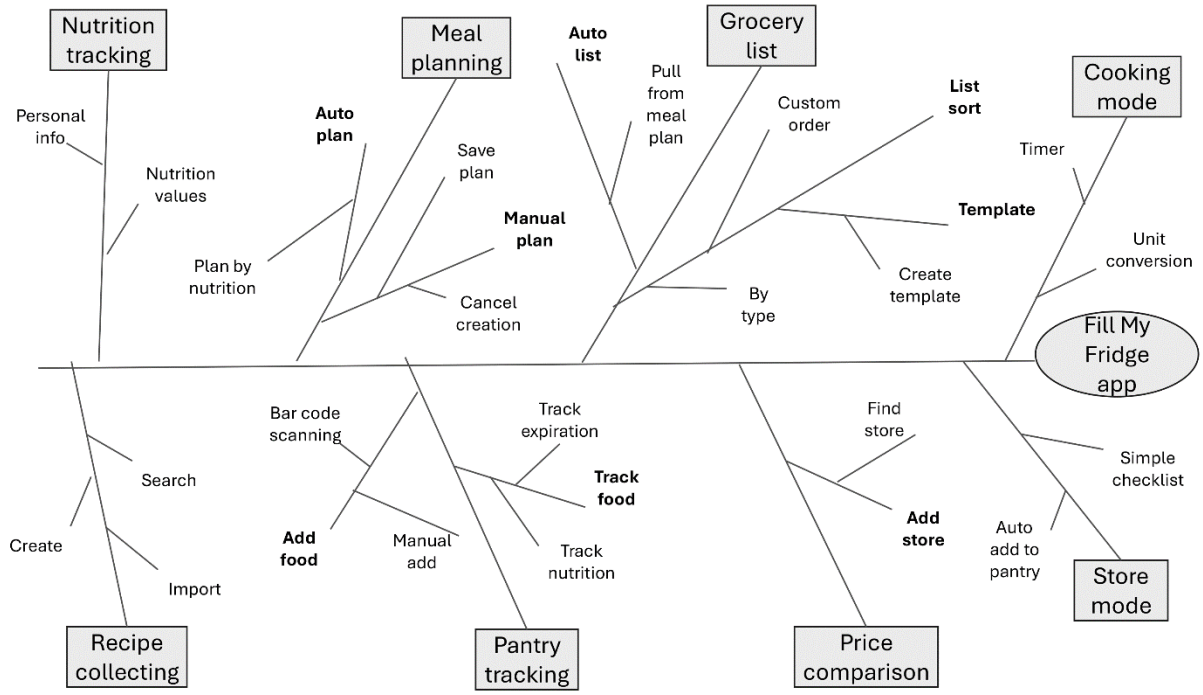


Figure 1: Partial feature tree for the Fill My Fridge app.

2.2. & 2.3. Scope of Initial Release & Subsequent Releases

Feature	Release 1	Release 2	Release 3

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FE-1 Nutrition Tracking	USDA Foundation Foods Imported. Users can view the nutrition of their recipes and meal plans. Users can set dietary restrictions	USDA Branded Foods Imported and Linked to Foundation Foods. Users can set macro nutrition goals to be compared against recipes.	USDA Nutrient Database Imported for Micronutrient Tracking. Users can set micro nutrition goals to be compared against recipes
FE-2 Recipe Collecting	Users can create recipes and map ingredients to foundation foods.	Users can publish recipes and browse recipes published by other users. Basic recipe site importing by detecting common html elements.	Recipes are automatically categorized based on ingredients, language used in steps, and common planned times to use recipe. Recipes can be published by subscribers
FE-3 Meal Planning	Users can create meal plans by linking recipes to calendar	Users can tell the app to generate a meal plan by gathering recipes that match the user's nutrition goals.	Meal plan can be shared by subscribers.
FE-4 Pantry Tracking	Not Implemented	Manual Importing for users to add what they have available prior to using the pantry tracker.	Automatically tracks expiration dates based on user's purchase history at stores.
FE-5 Smart Grocery List	Not Implemented	Meal plan can send a list of ingredients based on shopping interval set by user. It communicates with pantry tracker to automatically check off anything the pantry says it has.	Grocery List uses common expirations and sales to notify the user for the optimal times to shop based on desired mealtimes. List is automatically updated as Pantry is updated. List can be shared by subscribers
FE-6 Grocery Store Querying	Not Implemented	Users can select preferred stores in their area and the app will show the top 3 results when searching for each ingredient on the grocery list for the user to select.	Store inventories are mapped to Foundation Foods. Users can select brand preferences. Substitute options will be shown if there is nothing suitable available for the item
FE-7 Cooking Mode	Recipe Steps Are Paginated	Recipe steps can have defined timers set by user and saved to the recipe for reuse.	The app will detect times and have timers ready for the user. It will detect what is assumed to be done prior to cooking (such as cutting

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			vegetables) and add that as the first step
FE-8 Store Mode	Not Implemented	Users can see a simple checklist and locations in the selected store of the items. As they check things off, the items are added to the pantry tracker.	Store List shows items in order based on optimal path and tells the user which aisle to go to next

2.4. Limitations and Exclusions

LI-1: Images Associated with Recipes/Ingredients

LI-2: Cooking Instructions/Glossary

LI-3: Recipe Ratings

LI-4: Interoperability with other Recipe Apps

LI-5: Social features related to recipes beyond an upload/download

LI-6: Website/Desktop Application

LI-7: Nutrition info from Restaurants

LI-8: User created nutrition information

LI-9: OCR Recipe Importing

EX-1: Will not prioritize sponsored content over dietary restrictions or nutrition goals set by the user

EX-2: Will not charge money for anything that can be done locally on the phone

EX-3: Will not have Ads unrelated to food/groceries

3. Business Context

3.1. Stakeholder Profiles

Stakeholder	*Major Value*	*Attitudes*	*Major Interests*	*Constraints*
College Student	Cheaper shopping experience	Only value the ability to see prices ahead of time and compare stores for cheaper price	Availability of items in stores, price comparison, finding new recipes	Student needs access to a kitchen and place to store food

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Private Chef	Optimizing Job	Value ability to manage multiple kitchens and what food is in each kitchen. Additionally cares about getting food requests and easily adding requested meal ingredients to grocery list	Planning meals, adding new recipes, grocery shopping list	Must easily fit into workflow
Committed Patrons	Cross referencing shopping list to see store has it	Being able to see that their favorite store has everything on their list.	Store items availability and cost comparison	Must be 100% sure that the store actually has all items they are looking for
Grocery Stores	Increased reach	Interested in sharing information about their cheap prices compared to their competitors. Will bring in more patrons	Availability of items in stores, price comparison	Constraints must have compatible API
Restaurant	Managing inventory/expiration dates	Chefs spend less time portioning and taking inventory of ingredients	Pantry tracking	Has easily trackable inventory
Recipe Websites	Increased site clicks	If their site is able to be scraped, they will get more visits from our users who find recipes then easily add them to our app	Recipe import from the websites	Data must be scrapable and site cannot need account
USDA	More users	USDA releases this database for free to increase nutritional knowledge and support research. They will appreciate more	Nutrition	None

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		individuals tracking nutrition		
Competing Apps	Improved understanding of app/market trends in this app space	See our features and our method of implementing features	Comparing and rating the features we have	Can implement our ideas
Food Manufacturers	Improved sales	If they can create and share recipes specific to their products our users will be more likely to buy their specific variant of some grocery item	Ability to upload their own recipes	Must make it easy for companies to share recipes and that they can brand their recipes
Cafeterias	Planning the meals	They want to track the nutritional value in the meals they serve as well as figure out portioning for making mass food orders	Meal planning, grocery list, nutrition	Meal planning needs to be easily scalable
Stay at home parent(s)	Meal planning and tracking food/expiration dates	General use cases of stocking kitchen, making food, tracking nutrition, and shopping	Saving and adding recipes, planning meals ahead, grocery shopping list, pantry tracking.	Time and grocery needs

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